

Industrial Document Intelligence Platform

Objective

Build a production-ready AI system that allows users to upload technical documents and ask questions using Retrieval-Augmented Generation (RAG).

Problem

Engineers and researchers spend significant time searching through manuals, reports, and technical documentation.

Traditional keyword search lacks semantic understanding.

Users

- Engineers
- Researchers
- Technical teams

Core Features

Document Upload

Supported formats:

- PDF
- DOCX
- TXT

Document Processing

- Text extraction
- Chunking
- Embedding generation
- Vector storage

Retrieval

- Semantic search
- Metadata filtering
- Citation tracking

Question Answering

- RAG-based answers
- Grounded responses
- Source citations

Multi-Agent Workflow

Agents:

- Retrieval Agent
- Reasoning Agent
- Verification Agent
- Citation Agent

Evaluation

Metrics:

- Faithfulness
- Context Recall
- Answer Relevancy

Tools:

- RAGAS

Technology Stack

Backend:

- FastAPI

LLM Framework:

- LangGraph

Vector Database:

- Qdrant

LLM:

- OpenAI

Deployment:

- Docker

Testing:

- Pytest

Code Quality:

- Ruff
- Black